

EXPECTED RETURNS ON CURRENCIES

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Basic Question

How are interest rates and exchange rates related?

- Covered interest rate parity (CIP) gives an answer
- $\frac{F_t}{S_t} = \frac{1+i_t}{1+i_t^*}$
- F_t is the forward exchange rate, S_t is the spot exchange rate, i_t is the domestic interest rate and i_t^* is the foreign interest rate
- Interest rates today vs. future exchange rates?
- $i_t - i_t^*$ versus $\frac{S_{t+1}}{S_t}$

Economic Reasoning

- Supply and demand story:
 - ▶ The Fed increases interest rates
 - ▶ Mexican investors want to buy USD, high-yielding bonds
 - ▶ Creates demand for USD, which drives up value of USD
- What's wrong with this story?
 - ▶ Isn't this too good to be true for these Mexican investors?
 - ▶ Unclear about returns
 - ▶ what else is going on?
 - ▶ Risk
- We need to think more seriously about currency returns

Returns on Foreign Exchange

- i_t = one-year riskless MXN interest rate (from date t to $t + 1$)
- i_t^* = one-year riskless USD interest rate (from date t to $t + 1$)
- S_t = MXN/USD spot exchange rate, date t

Suppose you are a Mexican investor.

- What's the return on 1-year MXN?
- What's the MXN-denominated return on 1-year USD?
- We call this the “uncovered” MXN return on a foreign bond

Currency Returns

The uncovered, MXN-denominated return to investing one dollar in a USD-denominated bond between time t and one-year hence, time $t + 1$, is:

$$\frac{(1 + i_t^*)S_{t+1}}{S_t}$$

where S_t is MXN/USD. It is crucial to note that this is a MXN-denominated return. The derivation goes as follows.

- Start with MXN 1 at time t and convert it to USD at today's exchange rate, S_t . This gives USD equal to $1/S_t$
- Deposit the USD in a riskless U.S. financial system. After one year you will have USD $(1 + i_t^*)/S_t$.
- Where 'riskless' means that you know the interest rate at time t , for sure, and there's no credit risk

Currency Returns

- Back into MXN at the time $t + 1$ exchange rate. This gives you MXN

$$\text{MXN } \frac{(1 + i_t^*)S_{t+1}}{S_t}$$

- Since we started with MXN 1, this is also the rate of return

What would make it not too good to be true?

- MXN-return on USD bond equals MXN-return on MXN bond.
 - ▶ Makes no sense. S_{t+1} is random as we do not know until $(t + 1)$.
- Instead, how about expected returns equal?

$$1 + i_t = E_t \frac{(1 + i_t^*)S_{t+1}}{S_t}$$

- This is called uncovered interest rate parity (UIP). Rewriting:

$$\frac{1 + i_t}{1 + i_t^*} = E_t \frac{S_{t+1}}{S_t}$$

Thus, high USD interest rates reflect expected depreciation of USD.

Expectations Hypothesis

- CIP: $\frac{1+i_t}{1+i_t^*} = \frac{F_t}{S_t}$
- UIP: $\frac{1+i_t}{1+i_t^*} = E_t \frac{S_{t+1}}{S_t}$
- Combined: $\frac{F_t}{S_t} = E_t \frac{S_{t+1}}{S_t}$
- $\Rightarrow F_t = E_t S_{t+1}$

Two ways to say the same thing: “interest rate differential equals expected depreciation” is the same as to say “forward rate equals expected future spot rate.”

Approximations

$$\frac{1 + i_t}{1 + i_t^*} = E_t \frac{S_{t+1}}{S_t}$$

Subtract 1 from both sides and use $1 + i_t^* \approx 1$ and ignoring Jensen's inequality

$$i_t - i_t^* \approx E_t \frac{S_{t+1} - S_t}{S_t}$$

“MXN minus foreign interest rate differential equals expected depreciation rate of MXN.”

Approximations

Take (natural) logarithms and use $\log(1 + i) \approx i$

- $\log(1 + i_t^*) \approx i_t^*$
- $\log\left(\frac{(1+i_t^*)S_{t+1}}{S_t}\right) \approx i_t^* + \log(S_{t+1}/S_t)$
- As before, UIP results from equalizing expected returns
- $i_t - i_t^* = E_t s_{t+1} - s_t$
- where lower-case letters denote logs

“Interest rate differential equals expected depreciation rate of MXN/USD.”

Summary: Uncovered Interest Rate Parity (UIP)

Uncovered interest rate parity: “expected returns are equal.” Different ways to express this are:

- Exact UIP: $\frac{1+i_t}{1+i_t^*} = E_t \frac{S_{t+1}}{S_t}$
- Approximate UIP: $i_t - i_t^* \approx E_t s_{t+1} - s_t$
- Forward rate expectations hypothesis (EH):
 - ▶ Exact: $F_t = E_t S_{t+1}$
 - ▶ Approximate: $\log(F_t) \equiv f_t = E_t s_{t+1}$

Evidence UIP

The standard uncovered interest rate parity (UIP), after taking logarithms and ignoring Jensen's inequality, is given by

$$E_t s_{t+h} - s_t = i_{h,t} - i_{h,t}^*, \quad (1)$$

where

- E_t denotes the expectation conditional in time $1 \leq t \leq T$;
- s_t is the exchange rate (in logs);
- $i_{h,t}$ denotes the domestic interest rates that matures in period h ;
- $i_{h,t}^*$ denotes the foreign interest rates that matures in period h .

Evidence UIP

Empirically this condition has been examined by regressing the (expected) change in the bilateral exchange rate on short-term deposit interest rate differential –the so-called Fama regression.

$$\hat{s}_{t+h} - s_t = \alpha + \beta(i_{h,t} - i_{h,t}^*) + e_{t+h}, \quad (2)$$

where $\hat{s}_{t+h}$ is an approximation of the expected value.

In this regression, the intercept and slope coefficients implied by the UIP condition are equal to zero and one.

There is a well-established literature that documents the failure of the uncovered interest parity (UIP) condition.

Evidence UIP

Most empirical studies, however, report that the estimated slope coefficient is either negative or smaller than one ([Froot and Thaler, 1990](#), [Bansal and Dahlquist, 2000](#), [Burnside et al., 2006](#), among others).

This finding has been labeled the UIP puzzle. Several avenues have been explored as possible explanations for this puzzle.

First, most empirical studies have examined the joint hypothesis of UIP and rational expectations ([Isard, 1996](#), [Chinn and Meredith, 2004](#), [Bussiere, et al. 2018](#)).

In this line, other studies have examined whether exchange rate expectations are rational.

Evidence UIP

There is ample evidence that suggests the failure of the unbiasedness hypothesis (whether exchange rate forecasts are unbiased)-as prediction errors are often negatively related to the interest differential so that estimated slopes are negative ([Meese and Rogoff, 1983](#), [Froot and Thaler, 1990](#), [Flood and Rose, 2002](#), among others).

Second, some studies have noted that the Fama regression equation could be omitting one or more explanatory variables.

If economic agents are risk-averse, then the Fama regression equation needs to include a time-varying risk premia term (non-standard preferences, [Verdelhan, 2010](#), [Lustig, et al. 2011](#); or introducing disaster risk, [Farhi and Gabaix, 2008](#)). The exclusion of such a variable might bias the slope coefficient downward.

Evidence UIP

Other variables that could affect the time-varying risk premia term are capital controls, exchange rate regime, inflation rate, and terms of trade ([Farhi and Werning, 2014](#)). Nevertheless, these same studies are unable to match other essential characteristics of the data -such as the evolution of the real exchange rate ([Engel, 2015](#)).

Third, other studies have explored the presence of non-linearities in the Fama regression equation.

The presence of non-linearities was initially justified as the result of transactions costs ([Hollifield and Upal, 1997](#)) or limits to speculation ([Sarno et al. 2006](#)).

Ex-post Estimation

Recall that $s_t \equiv \log S_t$ UIP is:

$$E_t s_{t+1} - s_t = i_t - i_t^*$$

Define the "forecast error" as ϵ_{t+1}

$$\epsilon_{t+1} = s_{t+1} - E_t s_{t+1}$$

Substitute

$$s_{t+1} - s_t = i_t - i_t^* + \epsilon_{t+1}$$

Consider the OLS regression. What does UIP imply about coefficients α and β ?

$$s_{t+1} - s_t = \alpha + \beta(i_t - i_t^*) + \epsilon_{t+1}$$

Answer: $\alpha = 0$ and $\beta = 1$

Carry Trade

The regression

$$s_{t+1} - s_t = \alpha + \beta(i_t - i_t^*) + \epsilon_{t+1}$$

$\beta < 0 \Rightarrow$ interest rate differentials are negatively correlated with depreciation on the MXN.

An investor who holds funds in a high yield currency not only benefits from higher yields, but also tends to benefit from an appreciation in the long currency.

This implies that there exist excess expected returns in currency prices.

Interpretation

First, what are currency excess returns

Carry trade (like money-market hedge):

- Borrow MXN 1, convert into USD, invest, repatriate at maturity.
- payoff:

$$\text{MXN} \left(\frac{(1 + i_t^*) S_{t+1}}{S_t} - (1 + i_t) \right)$$

Forward trade (like forward hedge):

- Sign contract to receive USD 1.0, deliver MXN F_t .
- Payoff:

$$\text{MXN} (1.0(S_{t+1} - F_t))$$

Currency risk premiums are the expected values of these things

Interpretation

Now, we've seen that, conditionally, UIP is robustly rejected. This means:

$$i_t - i_t^* = f_t - s_t = \underbrace{E_t(s_{t+1} - s_t)}_{q_t} + \underbrace{f_t - E_t(s_{t+1})}_{p_t}$$

- $f_t \equiv \log F_t, s_t \equiv \log S_t$
- $q_t \equiv$ expected depreciation rate
- $p_t \equiv$ risk premium
- UIP implies $p_t = 0$ The regression evidence, in contrast, indicates $p_t \neq 0$. Suggests that currency risk matters.

Forward Rate

Note: the evidence necessarily means that

$$F_t \neq E_t S_{t+1}$$

- Forward rate is not an unbiased predictor of the future spot rate
- Similar evidence holds for interest rates, commodities
- Expectations hypothesis relies on risk neutrality.

Prediction

Above we have the carry-trade payoffs. The trick is to know when they are expected to be positive

Simplest approach: go long the high rate currency (buying).

- Inadequate. Need to go long when interest rate difference is higher than average
- Example: JPY rates are always low

Standard approach

- Use regression to predict the above excess returns

Predictive Regression

The standard UIP regression:

$$s_{t+1} - s_t = \alpha + \beta(i_t - i_t^*) + \epsilon_{t+1}$$

Invoke CIP

$$s_{t+1} - s_t = \alpha + \beta(f_t - s_t) + \epsilon_{t+1}$$

Algebra

$$s_{t+1} - s_t - (f_t - s_t) = \alpha + (\beta - 1)(f_t - s_t) + \epsilon_{t+1}$$

$$s_{t+1} - f_t = \alpha + (\beta - 1)(f_t - s_t) + \epsilon_{t+1}$$

UIP implies $\beta - 1 = 0$: "unpredictable excess returns to currency speculations (risk premiums)". We've found that $\beta - 1 < -2$

Implication: Can use the regression to build a trading strategy.

Carry Trading Strategy

Run the regression:

$$s_{t+1} - f_t = \alpha + (\beta - 1)(f_t - s_t) + \epsilon_{t+1}$$

obtain estimates of α and $(\beta-1)$

Each month, observe $f_t - s_t$ and compute the predicted excess return:

$$x_t \equiv E_t(s_{t+1} - f_t) = \alpha + \beta(f_t - s_t)$$

If $x > 0$ go long USD, short MXN (buying USD and selling MXN). If $x_t < 0$ do the opposite.

References

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